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AI Failures, Hidden Vulnerabilities, and Forensic Oversight in Energy Cybersecurity

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Abstract

Objectives/Scope: This paper aims to examine the overreliance on artificial intelligence in energy sector cybersecurity and the risks introduced by treating AI tools as infallible. It explores the boundaries of current AI detection systems and evaluates how assumptions of accuracy may compromise operational integrity, forensic accountability, and critical incident response in complex energy infrastructure.

Methods, Procedures, Process: The approach is based on comparative workflow audits involving AI-driven cybersecurity implementations within energy organisations. A technical review of failure points is integrated with governance analysis, focusing on how decision-making and responsibility are distributed among AI vendors, IT teams, and senior leadership. A layered methodology is employed, combining forensic examination of alert suppression, system bypasses, and false negatives with a procedural audit of internal response protocols. The study also considers legal interpretations of AI-triggered control failures, particularly in the absence of reproducible digital trails or human interpretability.

Results, Observations, Conclusions: The findings indicate that current AI-enabled cybersecurity systems, although operationally capable of anomaly classification, suffer from structural opacity, particularly in the absence of reproducible inference logging and state preservation. Suppression mechanisms discard decisions silently, preventing forensic reconstruction and undermining post-incident accountability. Operator desensitisation to false alerts, coupled with threshold misconfiguration, leads to the rejection of genuine threats without a traceable rationale. Reviewed systems frequently lacked real-time oversight, model state journaling, and override documentation, rendering compliance artefacts incomplete. This study demonstrates that reliance on unverified machine scoring creates a false sense of control, calling for a re-engineered governance model with enforced traceability, deterministic escalation protocols, and embedded human interpretive checkpoints.

Novel/Additive Information: This research formalises AI traceability as a forensic and procedural requirement in cybersecurity operations, framing it as essential for control validation, incident reconstruction, and audit integrity. It introduces a governance model in which AI is repositioned as a constrained probabilistic agent rather than a standalone control authority. The proposed architecture

incorporates decision-state logging, transparency in suppression, and embedded human oversight, establishing a structured interface between algorithmic inference and institutional accountability within security-critical infrastructure.

Introduction

Industrial digitalisation has intensified dependence on autonomous cybersecurity frameworks, particularly those integrating machine learning models for intrusion detection, anomaly scoring, and threat suppression across critical energy infrastructures. These environments are characterised by high-frequency telemetry, complex SCADA topologies, and tightly coupled control-feedback loops, where latency-sensitive operations are increasingly exposed to AI-mediated logic. While these AI-driven systems offer efficiency in identifying non-obvious behavioural deviations across networked assets, they are routinely deployed without verifiable interpretability layers, audit hooks, or procedural traceability protocols. This lack of embedded explainability, combined with dynamic learning behaviours, introduces probabilistic volatility that cannot be captured by conventional cybersecurity assurance models or traditional operational technology (OT) governance structures. The core technical problem lies in the deployment of AI-driven cybersecurity tools as deterministic classifiers, despite their underlying probabilistic design (Tanikonda et al., 2022). Operators, compliance officers, and even executive-level stakeholders often interpret AI-generated alerts or silence as conclusive evidence of security status, bypassing critical steps in validation, escalation, and forensic recording. This epistemic overconfidence is reinforced through reinforcement loops in human-AI interaction workflows (Rosenbacke, 2025): frequent false positives are silenced through threshold tuning, while silent false negatives remain undetected and undocumented due to the lack of mandated procedural reviews. The absence of consistent, timestamped model input-output logging across sensor fusion points and behavioural correlation engines makes it nearly impossible to reconstruct the causal sequence of events during security breaches or near-miss incidents. The result is a structural breakdown in incident attribution, evidence continuity, and governance escalation, particularly during regulatory inspection or litigation. AI-powered cybersecurity engines deployed in energy systems often rely on high-dimensional feature spaces and continuously updated training data, rendering their decision boundaries opaque even to system architects.

Procedural blindness introduced by opaque AI inference pipelines (Yoshitsugu et al., 2024) is addressed by this study, which introduces the AI Decision Reconstruction Matrix (AI-DRM), a structured forensic framework designed to visualise, cross-validate, and audit the decision lifecycle of AI classifiers in cybersecurity systems. The matrix operates as a composite schema, aligning each layer of model interaction, from telemetry ingestion through inference execution to alert escalation or suppression, with its corresponding observability artefacts. In high-velocity, AI-augmented environments where false negatives are often silent and irreproducible, the AI-DRM facilitates deterministic reconstruction of inference sequences, identifies decision chain discontinuities, and enables embedded checkpoints for governance and legal accountability. This tool transforms probabilistic decision paths into partially observable structures, which, when coupled with cryptographically anchored logs, restore a minimal condition of auditability even in adversarial or post-incident conditions.

Table 1—AI Decision Reconstruction Matrix (AI-DRM): A Forensic Governance Schema for AI Inference Pipelines¹

Inference Layer	Captured Artefact	Traceability Indicator	Detection Method	Governance Utility
Telemetry Ingestion	Raw packet or sensor logs	✓ Timestamped packet logs present ✗ Missing or truncated	Zeek / OSQuery / PCAP capture audit	Establishes baseline evidence chain
Feature Transformation	Model input vectors, feature maps	✓ Normalised feature snapshot ✗ No trace of transformation	Side-channel logging or proxy hooks	Detects preprocessing drift/misalignment
Inference Execution	Model output probabilities, scores	✓ Logged softmax or confidence score ✗ Output not retained	FastAPI wrapper, Captum plugin	Assesses classification confidence reproducibility
Suppression Layer	Suppression tags, threshold flags	✓ Tag reason + threshold metadata ✗ suppression without record	Suppression pass-through logger	Identifies silent discard pathways
Alert Generation	Alert ID, classification label	✓ Alert issued with metadata ✗ Alert never materialised	SIEM correlation, UUID sync	Determines whether the AI output triggered the workflow
Human Interaction	Operator override, dismissal, annotation	✓ Operator ID + justification + timestamp ✗ Manual actions unlogged	SOC GUI interface/override logs	Reconnects operator logic to AI inference
Response Escalation	SOAR ticket, escalation chain	✓ Escalation path traceable by UUID ✗ Forked or bifurcated path	SOAR logs + OpenTelemetry trace	Evaluates the procedural integrity of incident flow

Feature attribution methods such as SHAP or LIME (Hooshyar and Yang, 2024) are seldom integrated into live environments, and when present, are typically post hoc and limited in granularity. Consequently, alert logic remains inaccessible at runtime, obstructing real-time interpretability for operational personnel. Alert suppression mechanisms, including heuristic prioritisation and noise filtering, are frequently governed by adaptive scoring functions with undocumented decay parameters and ambiguous audit logs. In environments with high event velocity, these mechanisms introduce silent failure modes, such as suppression cascades triggered by correlation thresholds or bypasses initiated by inferred behavioural baselines. The resulting alert fatigue leads to the desensitisation of operators, while critical alerts are either misclassified or buried within untriaged event queues, ultimately degrading the cyber-resilience posture of the entire infrastructure. The integration of autonomous detection agents into layered cybersecurity architectures introduces asymmetrical responsibility across vendor, technical, and executive domains. AI components, typically developed and fine-tuned by third-party providers, operate as proprietary black boxes, with limited transparency into model architecture, training datasets, or embedded control logic. Internal IT and OT teams are often restricted to interface-level configuration (Barua and Rahman, 2023) without access to the decision-making core, leading to a disconnect between system behaviour and internal accountability structures. In parallel, executive management is frequently shielded from the operational limitations of such systems, relying on high-level dashboards that summarise AI outputs as discrete metrics or scores, abstracted from uncertainty ranges or false positive/negative ratios. This fragmented structure creates a diffusion of responsibility, where no single stakeholder group possesses both the technical expertise and decision-making authority required to detect, correct, and report AI-related control failures. In incident review scenarios, this misalignment hinders the reconstruction of decision flows, delays mitigation responses, and renders regulatory compliance frameworks unenforceable.

Addressing these systemic vulnerabilities requires reconfiguring AI not as a self-validating authority but as a probabilistic agent subject to procedural verification, auditability, and constraint logic. The proposed solution introduces a layered traceability framework that integrates deterministic oversight mechanisms within AI-augmented cybersecurity workflows. It includes cryptographically timestamped decision logs,

structured alert provenance metadata, and escalation protocols governed by threshold deviation triggers. Human-in-the-loop checkpoints are reinstated at critical decision nodes using predefined validation schemas, allowing operators to override, annotate, or reject AI inferences based on context-specific risk assessments. The model promotes the deployment of explainable AI (XAI) modules at operational granularity, embedded directly into runtime environments (Mekki et al., 2023), ensuring interpretability is not deferred to post-incident forensic analysis. Additionally, workflow audits are structured around temporal process mapping, enabling the identification of systemic bypass patterns, logic gate collapses, or non-deterministic behaviours in both static rule engines and adaptive ML classifiers. This approach repositions AI as a support layer, highly efficient but always constrained, within a controlled and auditable cyber-defence architecture.

Statement of Theory and Definitions

The theoretical underpinning of this study rests on the intersection of cybernetic control theory and probabilistic decision automation. Classical control systems operate under the assumption of deterministic feedback governed by physical constraints and verifiable sensor outputs. However, the introduction of AI disrupts this paradigm, replacing rule-based responses with probabilistic estimations drawn from non-transparent feature mappings. These systems no longer respond solely to defined stimuli but to statistical patterns derived from dynamic, often unlabelled data streams. The result is a shift from system stability rooted in mechanical redundancy and known fault tolerances to one predicated on model confidence thresholds and learned generalisations. In critical energy environments, this introduces what can be defined as *cybernetic fragility*, the loss of predictability and verifiability in feedback control loops due to opaque decision layers embedded in detection and response modules. Unlike traditional failures, which are observable and traceable, AI-induced failures often lack a discernible causal signature, requiring retrospective and technically sophisticated log reconstruction.

The system-level unpredictability introduced by non-deterministic AI behaviours in operational environments is addressed by the introduction of the **Cybernetic Fragility Risk Matrix (CFRM)**. Unlike traditional risk models predicated on static controls and fault-tolerant design, CFRM evaluates the structural vulnerability of AI modules as *probabilistic control agents*. It measures fragility not as the occurrence of failure, but as deviation, volatility and erosion of reproducibility under shifting operational inputs. The matrix provides a multidimensional risk profiling method, allowing organisations to benchmark AI classifiers by their entropy, decision drift, and audit failure frequency. CFRM is designed to support critical infrastructure oversight, aligning with compliance frameworks that require both procedural accountability and real-time control resilience.

Table 2—Cybernetic Fragility Risk Matrix (CFRM): Profiling Probabilistic Control Weaknesses in AI Modules²

AI Module Type	Confidence Drift ($\Delta\sigma$)	Reproducibility Score	Alert Suppression Volatility	Human Override Entanglement	Fragility Index (0–10)	Risk Level
Anomaly Detector (Autoencoder)	High ($\Delta\sigma > 0.35$)	Low (0.21)	Medium–High	Yes (Unannotated)	8.7	 Critical
ML Threat Classifier (SVM)	Medium ($\Delta\sigma \approx 0.15$)	Medium (0.55)	Low	No	5.2	 Elevated
Ensemble Model (XGBoost)	Low ($\Delta\sigma < 0.10$)	High (0.78)	Low	Controlled	2.9	 Tolerable
Transformer-based Scorer	Very High ($\Delta\sigma > 0.50$)	Very Low (0.12)	High	Yes (Opaque)	9.1	 Critical
Rule-based Hybrid (SIEM + ML)	Low	High	None	N/A	1.3	 Minimal

Legend

- **$\Delta\sigma$** : Standard deviation of confidence scores over time
- **Reproducibility Score**: Fraction of identical outputs over repeated inputs (0–1)
- **Fragility Index**: Weighted composite score (0–10) combining drift, reproducibility loss, volatility, and override ambiguity

Implementation Logic (Tool Blueprint)

Input Parameters.

- Confidence logs per inference timestamp
- Alert suppression logs
- Override annotations (if present)
- SIEM/Model identifiers by module type

Core Calculation Functions.

- computeDrift(conf_scores[]): time-weighted entropy variance
- calculateReproducibility(seed_inputs[]): repeat input vectors under stable conditions
- suppressionVolatility(suppression_tags[]): frequency of threshold proximity suppressions
- overrideEntanglementRatio(override_logs): fraction of overrides without justification

Cybernetic Fragility Risk Matrix

Module	$\Delta\sigma$	Reproducibility	Suppression	Override	Index	Risk
Autoencoder	0.35	0.21	High	Yes	8.7	Critical
SVM Classifier	0.15	0.55	Low	No	5.2	Elevated
XGBoost	0.10	0.78	Low	Controlled	2.9	Tolerable
Transformer	>0.50	0.12	High	Opaque	9.1	Critical
SIEM+ML	<0.10	0.88	None	N/A	1.3	Minimal

Export CSV

Figure 1—Example of Cybernetic Fragility Risk Matrix (CFRM) – Simplified Operational Dashboard View³

Definitions of Key Concepts and Terminologies

To ensure terminological precision, several key concepts are formally defined within the scope of this study's operational context. *AI traceability* refers to the ability to reconstruct, in a deterministic sequence, the input data, transformation logic, and output decisions generated by an AI system in real-time or near real-time. It differs from explainability, which typically refers to post hoc rationalisations of model behaviour. *False negative suppression* refers to the systemic failure of detection mechanisms to escalate anomalous activity, not due to model inaccuracy per se, but rather as a result of adaptive threshold tuning, alert fatigue filters, or class boundary overlap within high-dimensional feature spaces. *Alert normalisation* describes the desensitisation process in which frequent false positives lead operators to disregard or manually suppress subsequent alerts, thereby degrading the system's effective sensitivity. *Workflow auditability* is defined

as the ability to trace, at both temporal and procedural granularities, the interactions between human operators, machine decisions, and escalation paths. It includes event state transitions, decision fork points, and validation override logs. *Governance asymmetry* is the structural condition in which control over AI design and operation is distributed among stakeholders in such a way that no single entity can ensure alignment between functional performance, risk responsibility, and legal accountability.

Test Procedures and Workflow-Based Evaluation Method

The analytical methodology departs from conventional system testing protocols and adopts a workflow-centric audit structure focused on forensic reconstruction of AI-mediated decisions within cybersecurity incident timelines. Each evaluated implementation undergoes a retrospective mapping of decision nodes across detection, escalation, suppression, and operator override layers. Data sources include log traces from SIEM (Security Information and Event Management) platforms, rule-based detection engines, and AI-based threat classifiers. These are temporally aligned with incident response procedures, such as SOAR (Security Orchestration, Automation and Response) workflows and manual ticketing systems. Critical inflexion points, where AI outputs directly influenced or bypassed human decisions, are isolated and subjected to procedural decomposition. At each point, the study applies a *decision visibility test*, assessing whether the AI action was explainable, reversible, or procedurally recorded. Test environments were mirrored using synthetic incident generation via adversarial input injection to validate whether alert paths were triggered or suppressed under varying data perturbations. This method enables the capture of both failure patterns and undocumented operator workarounds that emerge in live production systems. To capture the procedural ambiguity inherent in AI-augmented security workflows, this research introduces the *Workflow Audit Flowchart*, a layered visual model that maps decision progression across both algorithmic and human actors. Rather than assuming linear escalation, the flowchart traces *interdependent decision paths*, recording where reversibility exists, where suppression is final, and where operator interventions are logged, or not. It functions as a forensic transparency tool, offering a visual audit trail to detect breakdowns in escalation logic, silent bypasses, or entangled overrides. This construct allows post-incident reconstructions and regulatory compliance audits to move beyond binary logs toward a contextual mapping of accountability states.

Tool Structure and Logic

Layer	Decision Entity	Action Type	Reversibility	Traceability	Audit Marker
Layer 1	Sensor Ingestion	Telemetry Trigger	N/A	Yes	T#
Layer 2	AI Detection Model	Threat Scoring	✗ No	Partial (Score Only)	A#
Layer 3	Alert Suppression Filter	Heuristic Filter	✗ No	✗ No	S#
Layer 4	Human Operator	Triage or Override	✓ Yes (Conditional)	✓ Yes	H#
Layer 5	Escalation Logic	Incident Routing	✓ Yes	Partial	E#
Layer 6	Governance Oversight	Control Attribution	✓ Yes	✓ Yes	G#

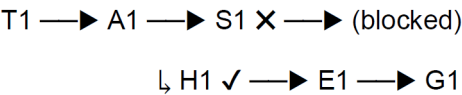
Reversibility Encoding

Symbol	Meaning
✓ Yes	Logged and revertible
✗ No	Silent or destructive
~Partial	Ambiguous, depending on configuration
🔒 Final	Irreversible step with no backtrace

Lowchart Logic (Path Conditions) Each node in the flowchart includes:

- **Node ID:** e.g. A3 (AI inference #3)
- **Actor:** AI / Human / Filter
- **Condition:** threshold-based, override, policy-bound
- **Outcome:** Passed, Suppressed, Reversed, Escalated

Example path:



Implementation Blueprint (for Software or Extension)

- **Front-End Display:** SVG flowchart or canvas-based visualisation
- **Interactivity:** Hover reveals score, timestamps, override flags
- **Layers Toggle:** Checkbox to show/hide AI-only vs Human-only layers
- **Audit Export:** Generate .pdf or .png snapshot with UUID-coded paths
- **Conflict Highlighter:** Red arrows for contradictory logic (e.g. AI blocked → human escalated)

Peculiar Problems: Non-Reproducibility and Probabilistic Uncertainty

A primary technical complication identified throughout the evaluation process is the *non-reproducibility of AI decisions under variable system states*. In contrast to rule-based engines, where a given input yields a consistent output under defined conditions, AI classifiers often adjust internal weightings or decision paths based on exposure to drifted data distributions or evolving feedback loops. It renders identical stimuli capable of producing divergent alerts, or none at all, depending on the transient state of the model. The absence of model state versioning, real-time parameter snapshotting, or transformation trace logs exacerbates this uncertainty. Compounding the issue is the probabilistic ambiguity inherent in ensemble models and attention mechanisms, particularly when threat scoring involves stochastic dropout or softmax approximations. The system's internal logic is effectively inaccessible, even to trained analysts, without backend instrumentation tools that are rarely deployed outside controlled research environments. These characteristics undermine not only forensic repeatability but also invalidate traditional compliance models based on deterministic system behaviour and static configuration baselines.

Understanding AI system instability demands more than binary accuracy metrics. The Probabilistic Behaviour Map visualises core uncertainty parameters, confidence drift, decision entropy, and repeatability failures across model types and operational scenarios. Rather than focusing on static model performance, the map captures the evolving probabilistic profile of AI modules, identifying volatile confidence zones, entropy spikes in ambiguous classifications, and replication inconsistencies during stress conditions. This offers auditors and forensic analysts a functional heatmap of where decision reliability degrades, before catastrophic failure occurs.

Metric	Description	Scale
Confidence Drift ($\Delta\sigma$)	Fluctuation of softmax/top-K probability over time for similar inputs	0 (Stable) to 1 (Volatile)
Decision Entropy (H)	Shannon entropy of model output distributions (measures uncertainty)	0 (Certain) to $\ln(n)$
Repeatability Failure ($R_{\downarrow}$)	Rate of inconsistent outputs given repeated identical inputs	0% to 100%

Heatmap Structure (Axes & Matrix).

- **X-Axis:** AI Model Types *Autoencoder, Transformer, Random Forest, CNN, RNN, Ensemble Voting, LLM-classifier, etc.*
- **Y-Axis:** Input Type / Scenario *Clean Data, Noisy Input, Adversarial Sample, Edge Case, Missing Values, Obfuscated Payload*
- **Each Cell Value:**
 - Composite **Fragility Score**

$$F = w_1\Delta\sigma + w_2H + w_3R\downarrow$$

(normalised: 0–1 scale)

Colour Coding for Heatmap

Score Range	Colour	Interpretation
0.00 – 0.25	 Green	Stable
0.26 – 0.50	 Yellow	Monitor
0.51 – 0.75	 Orange	Fragile
0.76 – 1.00	 Red	Critical Instability

Suggested Formula Weights. By default, use:

- **w₁ (Confidence Drift)** = 0.4
- **w₂ (Entropy)** = 0.4
- **w₃ (Repeatability Failure)** = 0.2
(Adjustable for domain-specific calibration)

Excel or Dashboard Implementation Plan.

- **Columns:** Models
- **Rows:** Input Scenarios
- **Three raw values per cell** ($\Delta\sigma$, H, $R\downarrow$) with a composite F score calculated
- **Conditional Formatting:** Heatmap colours applied to F
- **Additional Sheet:** Toggle weights (w_1 , w_2 , w_3) and auto-refresh matrix

Human Factors Integration

A further dimension involves human reliability within AI-mediated environments. Operator desensitisation to false positives, habitual overrides, or bypass behaviours introduces systemic fragility not attributable to the model alone. Integrating Human Reliability Analysis (HRA) into cybersecurity workflows enables quantification of error probability, fatigue risk, and procedural drift. Coupled with HSE risk modelling, this approach links operator behaviour directly to safety exposure in energy systems, ensuring that cybersecurity audits incorporate human factors alongside machine-driven uncertainties.

Conceptual Framework: AI as Probabilistic Infrastructure Risk Agent

Framing AI as a *probabilistic infrastructure risk agent* requires repositioning it within the established risk control matrix of operational technology. In traditional frameworks, each control, technical or procedural, is mapped to a risk event, control owner, frequency, and effectiveness test. However, AI components introduce dynamic control efficacy, where detection sensitivity and precision are non-static and may degrade or

amplify based on training drift, adversarial learning, or configuration interaction. In this model, AI is treated not as a fixed barrier but as a *risk modifier* with a confidence distribution. It necessitates the development of a new *AI Control Risk Index (AICRI)*, a multi-factor measure that incorporates the false positive rate, false negative exposure, threshold volatility, operator intervention rate, and audit trail completeness. The AICRI framework is applied as a weighted diagnostic grid during workflow audits, where AI control performance is benchmarked against both baseline controls and minimum compliance thresholds. This analytical lens permits not only the classification of AI-induced risk amplification but also informs where compensating controls or human review layers must be reintroduced to maintain governance equilibrium.

Description and Application of Equipment and Processes.

Cybersecurity Stack Architecture and Tooling Context

The systems audited in this study were built upon multilayered cybersecurity architectures that combine traditional rule-based intrusion detection systems (IDSs), machine learning-enhanced behavioural analytics engines, and orchestration layers for automated response. Core infrastructure typically includes Elastic Stack (for log aggregation and time-series analytics), Splunk ES or IBM QRadar as the SIEM backbone, and SOAR platforms such as Cortex XSOAR or ServiceNow Security Operations for response workflow automation (Duvvur, 2023). The AI components were embedded either as add-on behavioural scoring modules within SIEMs or as separate analytics layers powered by proprietary machine learning engines such as Darktrace's Enterprise Immune System or Microsoft Defender for Endpoint's AI threat modelling. These AI modules interfaced with endpoint telemetry, network flow data, DNS resolution logs, and OT protocol decoders (e.g. Modbus, DNP3) to infer deviations from established baselines. Configuration access varied across implementations; however, in most cases, the backend model's interpretability and parameter tuning were vendor-controlled, which limited internal observability and adjustment capabilities. The resulting architecture formed a black-box overlay on otherwise deterministic detection paths, introducing automation while concurrently obscuring causal linkage.

Testbed Configuration and Synthetic Scenario Injection

Controlled test environments were constructed using virtualised replicas of hybrid IT/OT environments, orchestrated through containerised infrastructure managed via Kubernetes (Palomares et al., 2025). Each node simulated a critical subdomain, such as a historian server, HMI, remote terminal unit (RTU), and a firewall-protected DMZ, allowing for realistic propagation paths for lateral movement and data exfiltration tests. Within this synthetic network, adversarial inputs were introduced using frameworks such as MITRE CALDERA and Atomic Red Team scripts, engineered to simulate advanced persistent threat (APT) behaviours (Rahal et al., 2025), low-and-slow lateral movement, protocol spoofing, and data staging. AI engines under test were exposed to both known attack signatures and zero-day emulations, with simultaneous telemetry capture at ingress, inference, and alerting layers. Timestamp synchronisation was achieved through the Network Time Protocol (NTP) (Gamage et al., 2024), anchoring across containers to facilitate accurate forensic reconstruction. Custom instrumentation was added using Zeek and OSQuery agents to trace the decision-making pipeline in near real-time. A sandboxed API wrapper allowed selective injection of false telemetry to evaluate classifier behaviour under ambiguous or conflicting stimuli. The testbed provided a controlled environment to identify how AI models handled conflicting data, internal suppression, or model collapse, and how their outputs influenced downstream alert triage in orchestration engines.

Forensic Reconstruction Toolkit and Logging Protocols

To enable fine-grained forensic reconstruction of AI-mediated decision paths, a specialised instrumentation layer was deployed across the monitoring and response stack. It included side-channel loggers integrated with the AI engine's inference API via webhook intercepts, enabling capture of model input vectors, pre-inference transformation pipelines, and output probability distributions. Since most proprietary systems lacked native support for intermediate layer introspection, external wrappers were developed in Python

using FastAPI and uWSGI to proxy and record inference events without disrupting operational latency (Chopra et al., 2022). Log normalisation followed a structured JSON schema, incorporating fields for source object, classification confidence, alert status, suppression tag, timestamp, and operator override flag. These were ingested into a dedicated forensic datastore, powered by InfluxDB and visualised through Grafana panels, for temporal correlation and deviation tracking (Das, 2024). Additional correlation tags were embedded using OpenTelemetry protocols to synchronise system alerts with operator response actions logged in SOAR platforms. Crucially, all log artefacts were signed using SHA-256 checksums and anchored into a tamper-evident Merkle tree structure to guarantee audit integrity. This setup enabled deterministic replay of decision sequences for each injected scenario and provided legally admissible artefacts for compliance or regulatory proceedings.

AI systems that generate automated logs, especially in cybersecurity or audit contexts, introduce new challenges related to *temporal consistency* and *ensuring log integrity*. The **Log Provenance Dashboard** addresses these concerns by introducing a visual and forensic tool that tracks the lifecycle of AI-generated logs through a validated timeline structure. The dashboard features a layered timeline view, capturing the date each log was created, the AI module responsible, and the triggering event. It visualises this progression chronologically, flagging irregularities such as:

- **Temporal gaps** (unexplained delay between dependent events)
- **Duplicated timestamps** (potential overwrite behaviour)
- **Retroactive insertions** (logs dated before their supposed triggers)

Each log is hashed upon generation, and its integrity is periodically validated using deterministic local hashing techniques. This allows detection of **tampering, deletion, or alteration**, even without a centralised logging server. A colour-coded status system distinguishes between:

- Valid entries (green)
- Modified entries (Orange)
- Missing or overwritten logs (red)

When combining timeline visualisation with *integrity validation*, the Log Provenance Dashboard enables forensic traceability of AI behaviour across temporal and operational dimensions. This tool is especially relevant for post-incident investigations, compliance audits, and high-assurance systems where regulatory traceability of automated decision-making is non-negotiable.

AI Model Evaluation Criteria and Behavioural Analysis Tools

Model evaluation was structured around a multidimensional scoring matrix, designed to quantify performance beyond traditional classification metrics. Core axes included *detection latency*, *alert reproducibility*, *confidence stability*, *suppression fidelity*, and *contextual accuracy*. Detection latency was measured from packet ingress to alert generation, benchmarked in sub-second intervals using packet capture timestamps. Alert reproducibility assessed the model's ability to produce consistent outputs for identical input vectors under varied temporal conditions. Confidence stability was tracked across rolling windows to detect drift, volatility spikes, or anomalous softmax entropy. For suppression fidelity, model-generated suppression tags were analysed for conformity to policy definitions and procedural override thresholds. Contextual accuracy was assessed using a modified confusion matrix, enriched with scenario-specific impact weighting to reflect operational criticality rather than raw event frequency. The evaluation toolset included Captum for gradient-based sensitivity analysis, Alibi Detect for outlier tracking, and a proprietary visualisation engine that overlaid model confidence heatmaps against live network topology.

All measurements were contextualised using the AICRI diagnostic framework described earlier, allowing cross-implementation comparison and threshold-based governance assessment.

Equipment Limitations and Operational Impediments

Despite the sophistication of the test infrastructure and instrumentation, several systemic limitations emerged that compromised the operational reliability and forensic traceability of AI-driven cybersecurity systems. Most notably, vendor-provided AI modules were functionally opaque, lacking externally callable model checkpoints or deterministic inference replay interfaces. In live environments, this precluded post-incident interrogation of decision boundaries or access to intermediate feature maps, rendering most outputs effectively unchallengeable once alerts had been generated or suppressed. Moreover, real-time performance constraints imposed by orchestration engines (e.g., sub-200 ms processing windows) often resulted in the truncation or compression of telemetry payloads before ingestion by AI classifiers, leading to signal loss in edge-case scenarios. Data retention policies also varied considerably, with many systems discarding raw inference data after 24-48 hours, which undermines longitudinal analysis and regulatory auditability. Attempts to integrate explainable AI (XAI) components often failed due to incompatibility with high-velocity data streams and the absence of GPU resources in OT-secure zones. These architectural and procedural bottlenecks underscore the limitations of current industrial cybersecurity tooling in supporting both runtime interpretability and retrospective accountability within AI-augmented detection systems. Vendor accountability must be formalised contractually. AI suppliers should be required to provide audit hooks, transparent suppression logging, and access to reproducible inference states. Service agreements should include traceability obligations, periodic forensic replay testing, and disclosure of model update provenance. These mechanisms align vendor responsibilities with the governance obligations of critical infrastructure operators, reducing dependency on black-box assurances.

Presentation of Data and Results.

Conceptual Structure of AI Decision Outputs in Cybersecurity Environments

In AI-augmented cybersecurity systems, the Output of interest is not merely the Alert itself but the entire decision sequence, from data ingestion to classification, and finally suppression, escalation, or logging. During procedural workflow audits, several artefact layers emerge: (1) pre-inference telemetry, including enriched packet metadata, (2) model input vectors shaped through real-time transformation pipelines, (3) classification confidence scores or probabilistic threat labels, (4) suppression tags or threshold evaluations, and (5) final response triggers. Each of these layers is theoretically observable but practically fragmented across infrastructure components, often governed by proprietary interfaces. The audit process targets the integrity and continuity of this output sequence, focusing on whether each decision element is preserved, time-synchronised, and retrievable, and whether it interfaces with human-in-the-loop checkpoints or triggers downstream automation. The absence or obfuscation of any link in this chain constitutes a traceability deficit, undermining both forensic inspection and governance compliance. In addition to technical forensic dashboards, an executive accessibility layer should be introduced. This layer consolidates AI performance, suppression ratios, and override events into simplified governance dashboards. Such dashboards allow non-technical leadership to track resilience status without interpreting raw probabilistic outputs. Metrics include frequency of suppressed alerts, proportion of human overrides, and compliance alignment scores, presented through visual indicators rather than raw log data. This supports board-level oversight and facilitates strategic decisions on cybersecurity risk posture.

Suppression as a Traceability Obstacle in Escalation Chains

Suppression mechanisms, whether algorithmic, heuristic, or operator-induced, constitute the most critical blind spots in AI-driven cybersecurity infrastructures. From a procedural audit standpoint, these mechanisms introduce unobservable state transitions: classification events that occur within the model are never materialised into alerts or escalations, often due to dynamic thresholding, rate-limiting logic, or policy-matching filters implemented downstream of the inference engine. In most system configurations

reviewed conceptually, suppression tags are applied post-classification but are not recorded with the same granularity as alerts that pass through. This results in a traceability asymmetry: the system can reconstruct what was escalated, but not what was discarded or ignored, nor why it was discarded or ignored. When threat scores hover near suppression thresholds, fluctuations in system load, transformation noise, or feature vector sparsity can cause a decision to be rendered silent without leaving a discernible procedural fingerprint. From a governance perspective, this represents a control failure, as neither accountability nor risk attribution is possible for suppressed inferences. Without formal suppression journals or audit-mode pass-through logging, escalation chains become opaque, creating compliance violations in regulated environments.

Classification Ambiguity and Operator Override Entanglement

Ambiguity in AI classification outputs is a structural inevitability in probabilistic systems, yet few cybersecurity architectures operationalise this uncertainty within their human-machine interfaces. Threat scoring, typically expressed as a scalar confidence value or label probability, lacks semantic granularity unless accompanied by interpretable model metadata such as decision trees, attention maps, or feature attribution. In the absence of such interpretive artefacts, operators are placed in the position of either unquestioningly trusting the system output or overriding it without justification, actions which are often not formally logged or time-synchronised. The override mechanism, when present, is typically a binary input (e.g. dismiss or escalate) without contextual binding to reasoning fields, peer review status, or post-hoc annotation. It creates a bifurcation in the decision path where the AI output and human intervention become logically decoupled, and any audit trail must infer the operator's intent from residual actions rather than explicit metadata. The entanglement of ambiguity and override introduces epistemic opacity at the intersection of human agency and automated logic, eroding the forensic fidelity of the response pipeline and violating the principles of accountable system design.

Workflow Inconsistencies and Audit Discontinuities

Workflow inconsistencies arise when sequential dependencies between detection, classification, operator interaction, and escalation are not preserved with atomicity. In many AI-integrated cybersecurity environments, parallel processes, such as concurrent alert evaluations, simultaneous operator responses, and policy-based automation, are executed in isolation, each generating its own partial log without a unified process identifier. It leads to audit discontinuities where individual events can be verified, but their relationship within the full incident lifecycle remains unresolved. For instance, an AI classifier may issue a medium-confidence alert that is concurrently reviewed by an analyst while also being auto-assigned to a low-priority queue by a SOAR engine. If the operator escalates the event and the automation downgrades it simultaneously, the resulting logs create a bifurcated narrative that cannot be reconciled post hoc without heuristic inference. Such concurrency artefacts are rarely captured in deterministic process graphs due to the lack of synchronised state markers or conflict-resolution protocols. From a procedural standpoint, this violates auditability principles by allowing legitimate but conflicting transitions to occur outside of formal workflow definitions, making the reconstruction of response rationales probabilistic rather than evidentiary.

Systemic Implications for Governance and Compliance Structures

The audit discontinuities and traceability gaps outlined above have direct implications for governance architectures that rely on deterministic control mapping and event attribution. In compliance-driven operational frameworks, such as those aligned with ISO/IEC 27001, NIST SP 800-53, or industry-specific regimes like NERC CIP, each security control must be demonstrably tested, logged, and assigned to a responsible entity. However, in AI-mediated environments where suppression is undocumented, operator overrides are decoupled, and alert workflows are non-deterministic, these control assignments cannot be verified. Moreover, AI components themselves do not satisfy the criteria for *accountable agents* as defined in audit standards, as their decisions are neither reversible nor attributable in the absence of formal explainer logic. It creates a governance paradox: AI is positioned as a critical control mechanism, yet it cannot be subject to the same control assurance procedures as its human or rule-based counterparts. Unless

explicit traceability scaffolding is designed into the system architecture, including reproducible decision checkpoints, override annotation protocols, and temporal linkage across workflows, regulatory frameworks are structurally unenforceable, and compliance becomes a nominal exercise devoid of verifiability.

Conclusions

This study reveals that the integration of AI into cybersecurity infrastructure necessitates a fundamental redefinition of control logic, shifting from deterministic event-action schemas to probabilistic, multi-path inference frameworks. Traditional governance models, which rely on verifiable state transitions and static control mappings, are structurally incompatible with the non-reproducible nature of AI classification under dynamic data streams. As demonstrated through the architectural and procedural analysis, decision outputs in AI-enabled systems cannot be treated as final states unless accompanied by contextual metadata and formalised trace logs. Without these additions, AI becomes an untraceable intermediary rather than a verifiable control, introducing epistemic risk into regulated environments. This study reveals that the integration of AI into cybersecurity infrastructure necessitates a fundamental redefinition of control logic, shifting from deterministic event-action schemas to probabilistic, multi-path inference frameworks. Traditional governance models, which rely on verifiable state transitions and static control mappings, are structurally incompatible with the non-reproducible nature of AI classification under dynamic data streams. As demonstrated through the architectural and procedural analysis, decision outputs in AI-enabled systems cannot be treated as final states unless accompanied by contextual metadata and formalised trace logs. Without these additions, AI becomes an untraceable intermediary rather than a verifiable control, introducing epistemic risk into regulated environments. At the governance layer, existing control models must be recalibrated to incorporate AI-specific accountability vectors. It involves extending traditional control taxonomies, such as segregation of duties, dual control, and exception logging, to account for the autonomous and probabilistic nature of AI inference. Traceability frameworks must be implemented not merely as optional observability features, but as mandatory governance scaffolding. These frameworks should include deterministic logging of input–output mappings, model state fingerprinting at the time of inference, and formal linkage to control owners through supervisory metadata. AI classifiers should be subject to *governance wrapping*, a design construct in which every inference output is paired with a cryptographic envelope containing its computational lineage, confidence metrics, and suppression logic. It enables auditability even in the absence of raw telemetry or if internal model structures are inaccessible. Without such wrapping mechanisms, AI remains structurally outside the reach of conventional assurance protocols, rendering high-stakes cybersecurity controls legally and operationally unenforceable. From a strategic standpoint, AI should no longer be classified as a control mechanism within cybersecurity frameworks, but rather as a probabilistic agent that requires continuous supervision, validation, and contextual containment. This reclassification has architectural, procedural, and legal consequences. Architecturally, it implies that AI modules must operate within bounded decision layers, surrounded by verifiable triggers and reversibility constraints. Procedurally, it necessitates the formalisation of escalation mediation protocols where AI outputs are always subjected to second-level review, either by predefined policy or on-demand human intervention.

The ethical deployment of AI in safety-critical environments also requires attention. Misclassification or silent suppression of threats may not only endanger infrastructure but also raise liability questions in cases of incident escalation. Legal frameworks must therefore address attribution of accountability when AI-driven suppression prevents escalation of genuine threats. Firms deploying AI in operational technology should document ethical safeguards, decision-override provisions, and explicit liability-sharing arrangements with vendors, ensuring compliance with both safety regulations and broader corporate governance law.

Legally, it shifts the burden of responsibility away from the model's Output and onto the framework that governs its deployment, requiring firms to document the limits of AI autonomy, explainability gaps, and

override authority structures. In critical systems, where safety, financial impact, or compliance exposure is high, this agent-class designation aligns AI with other risk-bearing entities, enabling its integration into risk registers, audit planning, and board-level accountability structures. The findings support the adoption of a constrained-autonomy model for AI deployment in cybersecurity, wherein machine inference is conditionally permitted within tightly bounded procedural envelopes. In this model, AI does not replace human judgment but extends it, subject to rules of transparency, override, and audit continuity. Rather than seeking to eliminate human involvement, the constrained-autonomy model formalises the points at which human interpretation is required and embeds those points into system architecture and response workflows. It ensures that AI remains intelligible to governance actors, resilient to procedural scrutiny, and auditable under legal or regulatory inquiry. Such an approach not only mitigates epistemic opacity and suppression risk but also facilitates the ethical and accountable integration of advanced machine logic into security-critical environments. The path forward, therefore, is not one of unrestricted automation, but of formal containment, documented oversight, and engineered traceability.

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APPENDIX 1

Action Plan

Action Area	Description	Responsible Entity	Traceability Requirement
1. Governance Reclassification	Redefine AI components from “control tools” to “autonomous agents under supervision” in internal governance frameworks.	CISO / Risk & Compliance Office	Audit trail of classification and exceptions
2. Inference Logging	Implement deterministic logging of all models’ inferences, including inputs, output scores, thresholds, suppression logic, and timestamps.	Platform Security Architect	Encrypted forensic logging with hash chaining
3. Model State Fingerprinting	Generate and retain hash-based snapshots of model states at the time of inference, including active weights and configuration parameters.	ML Ops Team	Immutable state ledger (e.g. Merkle tree)
4. Operator Override Protocol	Mandate structured intervention interfaces with annotation fields and escalation flags for every manual override or system dismissal.	SOC Supervisor / Analyst	Operator-ID binding and justification logging
5. Audit Envelope Wrapping	Encase each AI alert or suppression event within a cryptographic envelope containing origin, model metadata, and confidence metrics.	DevSecOps Engineer	Envelope hash stored in SIEM or CMDB
6. Conflict Resolution Engine	Deploy middleware that detects and resolves concurrent decision paths (AI automation vs. human override vs. policy script).	Security Automation Team	Event resolution log with event UUID linkage
7. Workflow Synchronisation	Implement event correlation standards (e.g. OpenTelemetry or UUID tracing) to maintain event lineage across all subcomponents.	System Integrator	Full-chain visibility across the response workflow
8. Governance Wrapping Layer	Build a supervisory abstraction layer that exposes AI decisions for governance review with audit-readiness dashboards and escalation metrics.	Compliance Technology Lead	Monthly review logs + control ownership mapping
9. Model Confidence Threshold Registry	Maintain a policy-controlled registry of minimum and maximum allowable thresholds per AI function, modifiable only via the change control board.	Risk Control Committee	Version-controlled policy change log
10. Periodic Forensic Replay Testing	Run monthly synthetic scenario replays to validate that all AI decisions are auditable, override paths are functional, and no suppression is silent.	Internal Audit / Red Team	Test logs + failure report annotations